Project Progress Report

Project-I (PROJ-CS-302A)

Date:

Project Title: EduScheduler: Intelligent academic scheduler

Project Group No.: G52

Supervisor's Name: Dr. Stuti Mehla

Name of Project Lab Teacher: Dr. Deepti Dhingra

Team Members:

Sr. No	Roll No.	Name	Project Lab Attendance (%)
1.	2823205	Ankit Kumar Sah	
2.	2823203	Yash	
3.	2823248	Vipul Gupta	
4.			

Functionalities of the System Completed:

The developed system provides a comprehensive set of functionalities as part of an integrated Academic ERP platform:

- User Authentication & Authorization
  Secure login using JWT with role-based access control for Admin, Teacher, and Student.
- Academic Management
  CRUD operations for Departments, Courses, Sections, and Classrooms along with faculty-course allocation.
- Timetable Management
  Centralized timetable creation, updating, and deletion with personalized views for teachers and students.
- Google Calendar Integration
  Automatic synchronization of timetable and meetings with Google Calendar, including Meet link generation.
- On-Demand Meeting & Broadcast System
  Teachers and Admins can create instant meetings and broadcast them to targeted audiences (All, Teachers, Students, Department, Section).
- Notification System
  Real-time alerts for meetings, timetable changes, announcements, and academic updates.
- Faculty Management
  Workload tracking, availability management, leave requests, and class swap functionality.
- Student Academic Tracking
  Attendance marking, assessment creation, grade entry, and performance analytics dashboards.
- Resource Management
  Booking and availability tracking of labs, classrooms, and equipment with conflict detection.
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- Meeting Management
  Role-based meeting visibility with “Join Meeting” functionality integrated into dashboards.

- Data Import/Export
  Bulk data upload using CSV for efficient data handling.
- AI Chat Assistant
  Provides automated assistance and answers to user queries within the system.

Sources used with References:

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   https://flask.palletsprojects.com/
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5. JWT.io. (2024). *JSON Web Token Introduction*.
   https://jwt.io/
6. SQLAlchemy Documentation. (2024). *SQLAlchemy ORM*.
   https://docs.sqlalchemy.org/
7. APScheduler Documentation. (2024). *Python Scheduler Library*.
   https://apscheduler.readthedocs.io/
8. Google OAuth 2.0 Documentation. (2024).
   https://developers.google.com/identity/protocols/oauth2

Additional Comments:

The developed system demonstrates a significant evolution from a basic timetable scheduler to a comprehensive Academic ERP platform integrating scheduling, communication, and resource management. Special emphasis has been placed on real-world applicability, ensuring that the system addresses practical challenges faced by educational institutions such as conflict-free scheduling, efficient communication, and centralized data management.

The system is designed with a modular and scalable architecture, making it easy to extend with future enhancements such as AI-based scheduling optimization, predictive analytics, and mobile-first features. Security considerations such as JWT-based authentication and secure handling of sensitive data have also been incorporated.

Signature of the Supervisor **Signature of the Project Lab Teacher**